Ex. No. 04	UTILIZE THE ASSETS AND MEDIA
Date of Exercise	
Git Hub Host Link	https://github.com/davii23/flutterlab.git

Aim

To add and display image, audio, and video assets in a Flutter application.

Description

This experiment demonstrates how to add media assets to a Flutter application. An image is stored inside the `assets/images` folder and configured in `pubspec.yaml`. The image is then displayed using `Image.asset()` and multiple images are arranged in a gallery using `GridView.count()`.

Procedure:

1. A new Flutter project named `media_app` was created using the Flutter command-line tool.
2. An assets directory was created inside the project to store media files. Separate folders such as images, audio, and video were created for organizing the media assets.
3. An image file named `flutter.png` was added to the `assets/images` directory.
4. The `pubspec.yaml` file was modified to register the image asset. Proper YAML indentation was maintained while specifying the asset path.
5. The `flutter pub get` command was executed to update the project and load the configured assets.
6. The `Image.asset()` widget was used in the Flutter application to load and display the image from the assets directory.
7. A `GridView.count()` widget was used to create an image gallery. The `crossAxisCount` property was set to **2**, allowing two images to be displayed in each row.
8. The application was executed using the Flutter run command and tested in the Chrome browser.
9. The configured image was successfully loaded and displayed as a gallery in the Flutter application.

Program

```
import 'package:flutter/material.dart';
import 'package:video_player/video_player.dart';

void main() {
  runApp(const MediaApp());
}
class MediaApp extends StatelessWidget {
  const MediaApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
```

```
      primarySwatch: Colors.indigo,
      useMaterial3: true,
    ),
    home: const MediaDemo(),
   );
  }
}
class MediaDemo extends StatefulWidget {
  const MediaDemo({super.key});
  @override
  State<MediaDemo> createState() => _MediaDemoState();
}
class _MediaDemoState extends State<MediaDemo> {
  late VideoPlayerController _controller;
  bool _hasVideoError = false;

  // CORS-friendly Wikimedia PNG image link for web compatibility
  final String _imageUrl =
      'https://upload.wikimedia.org/wikipedia/commons/1/17/Google-flutter-logo.png';
  // Public sample video stream
  final String _videoUrl =
      'https://flutter.github.io/assets-for-api-docs/assets/videos/bee.mp4';
  @override
  void initState() {
    super.initState();
    _controller = VideoPlayerController.networkUrl(
      Uri.parse(_videoUrl),
    );
    _controller.initialize().then((_) {
      setState(() {});
    }).catchError((error) {
      setState(() {
        _hasVideoError = true;
      });
    });
  }
  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Flutter Assets and Media'),
```

```
  backgroundColor: Colors.indigo,
  foregroundColor: Colors.white,
  centerTitle: true,
),
body: SingleChildScrollView(
  child: Column(
    children: [
      // Student details banner header
      Container(
        width: double.infinity,
        color: Colors.indigo.shade50,
        padding: const EdgeInsets.symmetric(vertical: 10, horizontal: 16),
        child: const Row(
          mainAxisAlignment: MainAxisAlignment.spaceBetween,
          children: [
            Text(
              'Name: David Paul P',
              style: TextStyle(fontWeight: FontWeight.bold, fontSize: 14),
            ),
            Text(
              'Reg No: URK23CS8005',
              style: TextStyle(
                fontWeight: FontWeight.bold,
                fontSize: 14,
                color: Colors.indigo,
              ),
            ),
          ],
        ),
      ),
      Padding(
        padding: const EdgeInsets.all(20),
        child: Column(
          children: [

            const Text(
              'Hello World',
              style: TextStyle(
                fontSize: 28,
                fontWeight: FontWeight.bold,
              ),
            ),
            const SizedBox(height: 25),

            // IMAGE SECTION
```

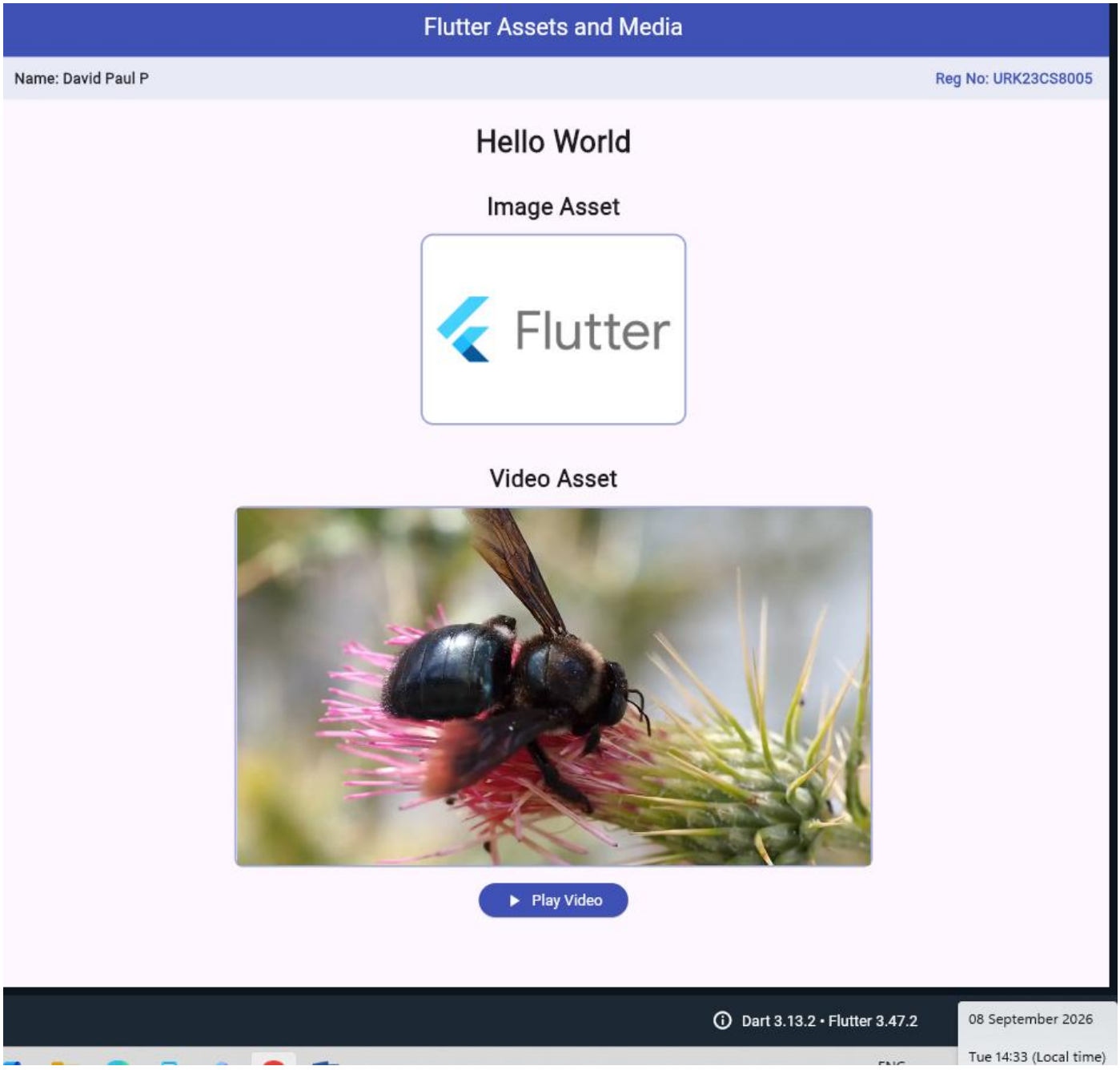
```
const Text(
  'Image Asset',
  style: TextStyle(
    fontSize: 22,
    fontWeight: FontWeight.bold,
  ),
),
const SizedBox(height: 10),
Container(
  width: 250,
  height: 180,
  decoration: BoxDecoration(
    color: Colors.white,
    border: Border.all(color: Colors.indigo.shade200, width: 2),
    borderRadius: BorderRadius.circular(12),
  ),
  padding: const EdgeInsets.all(12),
  child: Image.network(
    _imageUrl,
    fit: BoxFit.contain,
    loadingBuilder: (context, child, progress) {
      if (progress == null) return child;
      return const Center(child: CircularProgressIndicator());
    },
    errorBuilder: (context, error, stackTrace) {
      // Safe UI Fallback if the browser blocks network images completely
      return Container(
        color: Colors.indigo.shade50,
        child: const Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(Icons.flutter_dash, size: 60, color: Colors.indigo),
            SizedBox(height: 8),
            Text('Flutter Asset Display', style: TextStyle(fontWeight: FontWeight.bold)),
          ],
        ),
      );
    },
  ),
),
const SizedBox(height: 35),
// VIDEO SECTION
const Text(
  'Video Asset',
  style: TextStyle(
    fontSize: 22,
```

```
        fontWeight: FontWeight.bold,
      ),
    ),
    const SizedBox(height: 10),
    _hasVideoError
        ? Container(
            height: 200,
            width: double.infinity,
            color: Colors.grey.shade200,
            child: const Center(
              child: Text('Video stream failed to load.'),
            ),
          )
        : _controller.value.isInitialized
            ? Column(
                children: [
                  Container(
                    constraints: const BoxConstraints(maxWidth: 600),
                    decoration: BoxDecoration(
                      border: Border.all(
                        color: Colors.indigo.shade200,
                        width: 2,
                      ),
                      borderRadius: BorderRadius.circular(8),
                    ),
                    clipBehavior: Clip.antiAlias,
                    child: AspectRatio(
                      aspectRatio: _controller.value.aspectRatio,
                      child: VideoPlayer(_controller),
                    ),
                  ),
                  const SizedBox(height: 15),
                  ElevatedButton.icon(
                    style: ElevatedButton.styleFrom(
                      backgroundColor: Colors.indigo,
                      foregroundColor: Colors.white,
                      padding: const EdgeInsets.symmetric(
                        horizontal: 24,
                        vertical: 12,
                      ),
                    ),
                    onPressed: () {
                      setState(() {
                        if (_controller.value.isPlaying) {
                          _controller.pause();
                        } else {
```

```
                        _controller.play();
                      }
                    });
                  },
                  icon: Icon(
                    _controller.value.isPlaying
                        ? Icons.pause
                        : Icons.play_arrow,
                  ),
                  label: Text(
                    _controller.value.isPlaying
                        ? 'Pause Video'
                        : 'Play Video',
                  ),
                ),
              ],
            )
          : const Padding(
              padding: EdgeInsets.all(20.0),
              child: CircularProgressIndicator(),
            ),
        ],
      ),
    ),
  ),
 ],
),
),
);
}
}
        ],

      ),
    ),
  );
```

Output



Result

Thus, the Flutter application was successfully developed to load and display image assets. The image was successfully displayed in a 2-column image gallery using `GridView.count()`.